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Bubble evolution and gas-liquid mixing mechanism in a static-mixer-based plug-flow reactor: A numerical analysis

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ABSTRACT

Effective gas-liquid mixing is critical for the successful implementation of ozonation and advanced ozone-based oxidation technologies. This study focuses on a static-mixer-based plug-flow reactor derived from the HiPOx^{TM} reactor system, which is divided into three parts. The investigation delves into the bubble evolution and gas-liquid mixing mechanism through Computational Fluid Dynamics simulations. A comparative analysis of the gas-liquid flow fields, with and without the static mixer, reveals that the mixing element primarily enhances the mixing rather than the gas dispersion. Furthermore, the study employs relative standard deviation (RSD) and bubble diameter to assess the degree of the gas-liquid mixing. The results indicate that the plug-flow reactor equipped with six mixing elements achieves the optimal mixing performance (RSD 0.793 and bubble diameter 1.384 mm) with the lowest energy consumption. Compared with the reactor with four elements, the RSD decreases by 25.7 % and the bubble diameter decreases by 23.0 %. Additionally, the energy consumption is 37.6 % lower than that of the reactor with eight elements. This investigation serves as a foundation for the application of static mixers in gas-liquid mixing and provides a basis for further research in this area.

1. Introduction

Ozonation and advanced ozone-based oxidation technologies [1–3] have found widespread application in contaminant removal by generating highly oxidizing reactive oxygen radicals (mainly hydroxyl radicals, HO•). These radicals promote the formation of more biodegradable intermediates and often fewer toxic ones, making them advantageous for implementation upstream of biological treatments. Advanced oxidation process [4] involves radical chains initiated by the reaction between ozone (O_3) and hydrogen peroxide (H_2O_2). The realization of this reaction depends on the diffusion of ozone from the gas to the liquid phase in a multiphase flow. Therefore, the mass transfer of ozone is a critical factor affecting the efficiency and energy consumption of wastewater treatment in reactors.

HiPOx^{TM} is a reactor technology [5] implemented for ozone homogenization and peroxone reactions [6] using a static mixer as the ozone contact element. Its multiple advantages, including precise O_3/H_2O_2 dosing and forced mixing, enable efficient and low-cost removal of refractory organic pollutants from water [7–9]. However, this reactor technology encounters significant challenges in the chemical industry when treating organic wastewater with a large treatment capacity, high organic concentration, and high salinity. More ozone needs injecting to achieve a high removal efficiency. Nevertheless, the limitation of the mixing capacity of the static mixer at a certain liquid-phase flow rate results in untimely dispersion and eventually the formation of gas accumulation in the multiphase reactor [10–12]. Correspondingly, the ozone utilization and organic removal rates decrease signifucantly [13].

A static mixer is an energy-efficient component widely employed in

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numerous homogenization processes. Many studies have been conducted on the flow resistance, residence time, pressure drop, bubble distribution, heat transfer and mixing performance in static mixer [10,11,14–20]. Unlike active mixers, static mixers do not contain moving parts. They enhance radial mixing and derive mechanical energy from the momentum of the flowing fluid, resulting in lower energy consumption and reduced maintenance. The fluid behavior in tubular reactors incorporating a static mixer approximates plug flow conditions, as evidenced by comparisons made in some studies regarding the mixing capabilities and residence time distributions between various static mixer-based reactors, hollow tube reactor and plug flow reactor [21–23]. In addition, the hydrodynamics within the static mixer play a crucial role in mass-transfer efficiency. Understanding and leveraging these dynamics is a key strategy for addressing uneven gas-phase dispersion in the gas-liquid flow of the reactor. Many studies have been conducted in this field, covering dispersive mixing from laminar to turbulent regimes to increase the interface between immiscible phases [24–29]. However, the intricate flow field inside a static mixer makes it challenging to rely solely on experimental investigations and theoretical analyses. Advanced flow field testing techniques such as chemical probes and Laser Doppler Velocimetry can provide local characteristics of the flow field in single-phase systems [28]; however, obtaining detailed information on the flow field in gas-liquid or multiphase systems remains challenging. Additionally, experimental measurements are susceptible to flow-field interference, limited measurement accuracy, high costs, and long cycles. With the development of flow field simulation technology, Computational Fluid Dynamics (CFD) numerical simulations have been increasingly applied in multiphase flow-field exploration owing to their convenient analysis of the flow velocity, turbulence, heat transfer, and multiphase mixing processes by utilizing computer-based simulations. To study the reduction in CO_2 emissions in the cement industry, Moon et al. [30] utilized CFD to describe the flow and mass transfer of CO_2 and reactants in a fluid, while considering the particle dynamics of solid reactants. Heniche et al. [31] conducted a numerical investigation to explore the influence of blade shape on static mixing performance. They provided insights into the flow characteristics and mixing efficiency, highlighting the usability of CFD for evaluating different design parameters for static mixers. Moreover, van Wageningen et al. [32] assessed various CFD methods for dynamic flow analysis in a Kenics static mixer and determined the importance of model selection and validation to ensure reliable CFD simulations. Despite progress in utilizing CFD for multiphase flow-field analysis in static mixers, certain limitations need addressing. These limitations include the complexity of modeling and parameter selection in CFD simulations owing to phenomena such as gas-liquid interfaces. Furthermore, simplifications or two-dimensional simulations are often used to overcome computational limitations, which may introduce discrepancies in the actual operating conditions.

The function of the mixing element in a static mixer is crucial for achieving a low pressure drop and effective mixing. Several studies have been conducted on the structure and number of the mixing elements. Nyande et al. [22] found that by increasing the gap between the mixing elements, the pressure drop could be effectively reduced without significantly affecting the mixing performance. Tajima et al. [33] analyzed CO_2 drop formation process in a static mixer and found that increasing the number of mixing elements at higher flow rates effectively reduced the size of the CO_2 drops. Demoz [34] used various numbers of static mixer elements to investigate the optimal mixing parameters for the dewatering performance of coagulated mature fine tailings. The results showed that as the number of elements increased, the flow rate at which the peak dewatering performance was achieved decreased. Hosseini et al. [35] studied four-blade static mixers with 6, 8, and 10 elements in a saltwater and water system. They found that the performance of 10 elements was the best, although good mixing performance could also be achieved with 8 elements. Rabha et al. [14] reported on the visualization and quantitative analysis of dispersion and mixing processes in vertical gas-liquid flows using a helical static mixer. The research revealed enhanced mixing efficiency and an altered gas-liquid interface owing to the helical structure, providing a solid foundation for design improvements, optimization of flow conditions, and validation through CFD simulations. Biard et al. [36] investigated the enhancement of the O_3/H_2O_2 advanced oxidation process using a static mixer. The results demonstrated that the static mixer significantly improved the effectiveness of the O_3/H_2O_2 reaction by enhancing the mixing and contact of the oxidants, thereby improving the degradation efficiency. Obviously, the application of static mixer in gas-liquid mixing is limited, and there is a restricted understanding of the mechanism of gas-liquid mixing in a static mixer.

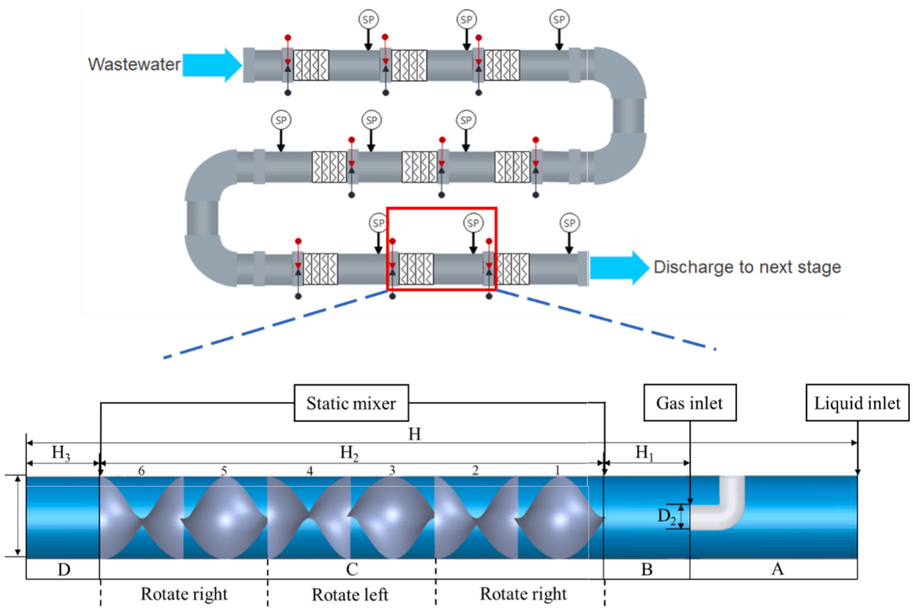
Ordinarily, the limitation of the gas-liquid mixing capacity in this HiPOx^{TM} reactor significantly reduces the ozone utilization rate and organic removal rate. Hence, analyzing the gas-liquid flow field and evaluating the effect mechanism of the static mixer on the gas-liquid mixing are crucial to solve the mixing imperfection and gas concentration gradients. Herein, the gas-liquid flow field in the plug-flow reactor with and without a static mixer was compared using CFD simulation to understand the flow pattern development of the bubble flow. The simulation results were first validated using previous experimental studies to ensure the accuracy of these CFD simulations. Based on this, a static mixer with six mixing elements was divided into three parts, each involving two mixing elements, to study the influence mechanism of the mixing elements at different positions on the gas dispersion and gas-liquid mixing. Subsequently, internal flow fields with varying numbers of mixing elements were used to achieve the best mixing capability and minimum energy consumption. This work provides a foundation for the application of a static mixer on gas-liquid mixing in future research.

2. Mathematic methods

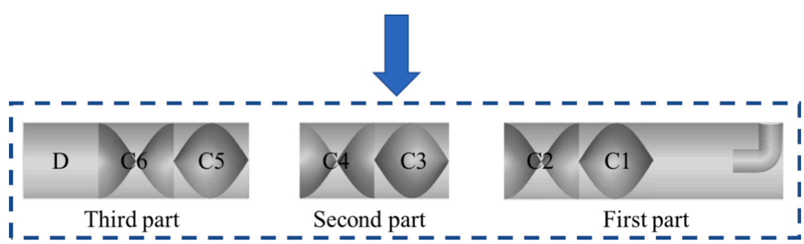
2.1. Geometry of static-mixer-based plug-flow reactor

A multiunit flow diagram of the HiPOx^{TM} system is illustrated in Fig. 1(a), comprising a sequence of injection-mixing-reaction modules designed to maintain a low instantaneous O_3/H_2O_2 molar ratio. In the peroxide process, wastewater treated with a specific amount of liquid hydrogen peroxide traverses the entire system and is mixed with ozone at varying dosages injected from the dosing point in each reactor unit. These two phases promptly pass through the static mixer element for ozone dispersion and liquid-gas mixing. After reacting for a short time, the primary treated wastewater advances to the next unit. For this study, one of the circulating elements in the HiPOx^{TM} system, featuring a gas injection port and a static mixer, was chosen, and its geometry is depicted in Fig. 1(a). It was divided into four zones: Zone A (liquid inlet zone), Zone B (gas inlet zone), Zone C (mixing zone) and Zone D (gas-liquid outlet zone); these structure parameters are listed in Table 1. The selection of the two-phase velocity was mainly based on the research described in Section 3.1. The liquid was injected from the liquid inlet in Zone A and flowed around the air inlet pipe to Zone B, entraining the gas phase injected at the right end of Zone B to Zone C. Six mixing elements were assembled in Zone C, where the first two elements, C1 and C2, rotated to the right; elements C3 and C4 rotated to the left; and elements C5 and C6 rotated to the right. Through the effect of the combination of left- and right-handed rotations, the liquid and gas were forcibly mixed by violent disturbances in the fluid, and the gas-liquid mixture flowed to Zone D.

To investigate the influence of the mixing elements at different positions on the gas dispersion and gas-liquid mixing, the static mixer with six mixing elements was divided into three parts, as shown in Fig. 1(b). The gas inlet pipes, C1 and C2, were in the first part, whereas elements C3 and C4 constituted the second part. Elements C5 and C6 and Zone D were in the third part. In this arrangement, the outlet velocity of the preceding part was set as the inlet velocity of the succeeding part.



(a) HiPOx™ system and a single unit



(b) Three parts of the static mixer

Fig.1. Schematic diagram of a plug-flow reactor intercepted from HiPOx™ system.

Table 1
Structure and operating parameters of static-mixer-based HiPOx™ reactor.

Parameter	
Total length H (mm)	822
Length of Zone B H_1 (mm)	85
Length of Zone C H_2 (mm)	480
Length of Zone D H_3 (mm)	80
Pipe diameter D_1 (mm)	80
Gas inlet diameter D_2 (mm)	24
Aspect ratio of element	1
Density of the primary phase (water) ρ_w (kg/m^3)	1000
Density of the secondary phase (air) ρ_a (kg/m^3)	1.225
Velocity of the primary phase (water) v_w (m/s)	0.5–1.7
Velocity of the secondary phase (air) v_a (m/s)	0.033–0.66

2.2. Mathematic model

The flow fields within this plug-flow reactor can be treated as a superposition of two levels: turbulence and liquid-gas multiphase flow. To realize the gas-dispersion process from the gas column to the bubble, the volume of fluid (VOF) was selected [37–39], which emphasizes the accurate capture of the phase interface between two or more immiscible phases.

The mass conservation of the phase was expressed by solving the continuity equation of the volume component of one or more phases. For the q^{th} phase, there was the following continuity Eq. (1),

$$\frac{\partial \alpha_q}{\partial t} + \vec{v} \cdot \nabla \alpha_q = \frac{S_{\alpha_q}}{\rho_q} \tag{1}$$

where α_q is the volume fraction of the q^{th} phase, which varies between 0 and 1, and $\vec{v}$ is the velocity.

A single momentum equation was solved throughout the domain, and the resulting velocity field was shared between the two phases. However, variations in velocity and volume fractions occurred between grids, leading to shear forces due to velocity gradients across grids and near wall surfaces [40]. Consequently, this resulted in the deformation and fragmentation of bubbles across the grids. This momentum equation was dependent on the volume fraction of fluid phase with the properties ρ and μ, given by

$$\frac{\partial}{\partial t}(\rho \vec{v}) + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + [\mu(\nabla \vec{v} + \nabla \vec{v}^T)] + \rho \vec{g} + \vec{F} \tag{2}$$

In general, for a q^{th} phase system, the volume fraction averaged density ρ took on the form

$$\rho = \sum \alpha_q \rho_q \tag{3}$$

All other properties (e.g., viscosity) were computed in the same manner. An implicit interface tracking-algorithm in the VOF was used to obtain a numerical solution.

The surface tension model used in this study was continuous surface force model [41]. It interprets surface tension as a continuous, three-dimensional effect across an interface, rather than as a boundary value condition at the interface. The surface tension effects were modelled by adding a source term to the momentum equation, that is, $\overrightarrow{F}$.

When passing through the gas-liquid interface, the pressure drop caused by surface tension was determined by the surface tension coefficient σ and the surface curvature of the orthogonal direction of the two radii.

$$p_2 - p_1 = \sigma\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \tag{4}$$

where p_1 and p_2 are the pressures on both sides of the two-phase fluid interface, and R_1 and R_2 are the principal curvature radii on the interface.

Let n be the surface normal vector, defined as the gradient α_q of the volume fraction of q^{th} phase, then

$$n = \nabla\alpha_q \tag{5}$$

The curvature k is defined in terms of the divergence of the unit normal $\widehat{n}$:

$$k = \nabla\cdot\widehat{n} \tag{6}$$

$$\widehat{n} = \frac{n}{|n|}$$

The force at the surface can be expressed as a volume force by using the divergence theorem. The volume force is the source term added to the momentum equation. It has the following form:

$$F_{vol} = \sum_{pairsij,i<j} \sigma_{ij} \frac{\alpha_i\rho_i k_j \nabla\alpha_j + \alpha_j\rho_j\kappa_i\nabla\alpha_i}{\frac{1}{2}(\rho_i + \rho_j)} \tag{7}$$

This expression allows for the smooth superposition of forces near cells in which more than two phases are present. If only two phases were present in a cell, then $k_i = -k_j$ and $\nabla\alpha_i = -\nabla\alpha_j$, and the above equation is simplified to

$$F_{vol} = \sigma_{ij}\frac{\rho\kappa_i\nabla\alpha_i}{\frac{1}{2}(\rho_i + \rho_j)} \tag{8}$$

where ρ is the volume-averaged density computed using by Equation (3). This equation shows that the surface tension source term for a cell is proportional to its average cell density.

Furthermore, the SST k-ω was used as the turbulence model, which is suitable for those with the shear effect on the wall. The SST k-ω model combines the advantages of the k-ε and k-ω models and provides more accurate modelling of turbulent behaviour and boundary layer flow in the near-wall region. Compared with other models, the SST k-ω model demonstrates stronger predictive capabilities in vortex regions and

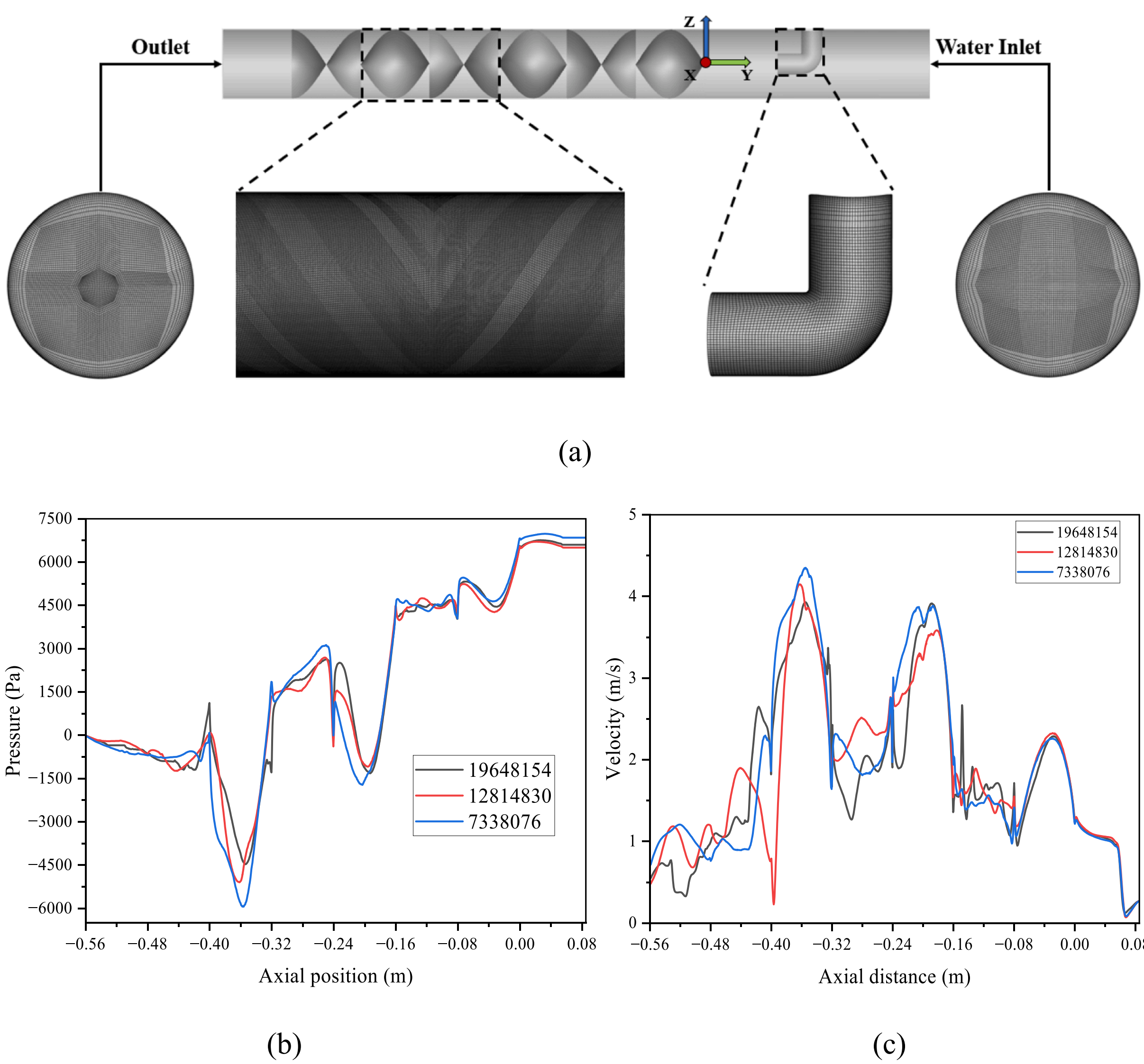


Fig.2. Mesh with static mixer (a) and grid independence analysis in the pressure (b) and velocity (c) distribution along the axial direction.

strong gradient flows. This model is effective in capturing the turbulent structures and vortex generation within a static mixer, thus allowing for accurate predictions of the flow fields and mixing effects [42,43]. The model consists of the turbulent kinetic energy and dissipation rate of the turbulent energy consumption of the element as follows:

$$\frac{\partial(\rho k)}{\partial t}+\frac{\partial(\rho k u_i)}{\partial x_i}=\frac{\partial}{\partial x_j}\left(H_k\frac{\partial k}{\partial x_j}\right)+F_k-Z_k+R_k \tag{9}$$

$$\frac{\partial(\rho\omega)}{\partial t}+\frac{\partial(\rho\omega u_i)}{\partial x_i}=\frac{\partial}{\partial_{x_j}}\left(H_\omega\frac{\partial_\omega}{\partial_{x_j}}\right)+F_\omega-Z_\omega+C_\omega+R_\omega \tag{10}$$

where H_k And H_ω represent the effective diffusivities of the turbulent kinetic energy and dissipation rate respectively. F_k and F_ω represent the corresponding generated phases, Z_k and Z_ω are the dissipation rates, C_ω is the diffusion term, R_k and R_ω are the user-defined source terms.

2.3. Simulation conditions

Fig. 2(a) illustrates the computational domain of the model, with the entire computational domain being segmented into structured hexahedral grids generated by ICEM CFD. ICEM CFD is a software package employed to create both 3D structured and unstructured meshes for CFD simulations. The complex critical flow areas were refined. Grid independence was analyzed by comparing different meshes (26,120, 148,479, 559,859, 3,879,488, 7,338,079, 12,814,830 and 19,648,154). The pressure distribution along the axial direction for varying grid numbers,7,338,076, 19,648,154 and 12,814,830 were presented in Fig. 2(b). As shown in Fig. 2(b), the 19,648,154 and 12,814,830 grids exhibited greater consistency compared to the 7,338,076 grids. Furthermore, the relative average error was used to quantitative analysis of grid independence, especially given the uncertainty in the velocity curves as shown in Fig. 2(c). The formula for calculating relative average error is provided below.

$$\eta=\frac{1}{n}\sum_{i=1}^{n}\left|\frac{x_i-\widehat{x}_i}{x_i}\right| \tag{11}$$

where x_i represents the velocity corresponding to the i^{th} data point of a certain number of grids, and $\widehat{x}_i$ is the predicted value. The maximum number of grids (19,648,154) was set as the predicted value. Upon calculations, the relative average error between 19,648,154 and 12,814,830 grids was 5.69 %, whereas that between 19,648,154 and 7,338,079 grids was 22.89 %. Therefore, grid numbers greater than 12,814,830 were used in this study.

The pressure-outlet boundary condition was used at the outlet, where the pressure was set to 1 atm. Two velocity-inlet boundary conditions were applied at the gas and liquid inlets. Table 1 listed the operating parameters used in the simulations. Specifically, four operating parameters corresponding to different flow patterns and subsequent simulations under a liquid velocity of 1.0 m/s with varying gas velocities (ranging from 0.033 to 0.231 m/s) were considered for model validation and bubble evaluation, respectively. Furthermore, an analysis of the gas-liquid mixing mechanism was performed under one of the actual working conditions during engineering operation, with a gas inlet velocity of 0.27 m/s and a liquid inlet velocity of 1.00 m/s. Water and air were selected as the liquid and gas, respectively, to simulate two-phase flow in the reactor unit. The simulations were conducted using ANSYS FLUENT 19.0 software. Herein, conservation equations were spatially integrated using a second-order upwind discretization scheme for convective terms, employing the PRESTO pressure interpolation scheme and adopting the coupled pressure–velocity coupling algorithm as the optimal numerical calculation method. The unsteady solver was used as the convergence strategy, with a time step of 5×10^{-4} s, and the accuracy of the convergence criterion was set to 1×10^{-4}. In addition, the extraction of the bubble size was compiled using User-defined Functions (UDF) in this simulation.

3. Results and discussion

3.1. Model validation and bubble evolution

In the case of the air-water multiphase flow, validation was conducted by comparing the simulation results with the experimental measurements from a previous study [44]. It became evident that the combination of the VOF multiphase model and SST *k-ω* provides a reliable prediction of the gas-liquid flow field (Fig. 3).

Four out of six flow patterns in Zone D of the reactor were compared in the experimental [44] and simulation results. These four typical flow patterns comprised three composite flow patterns, namely stratified bubble flow ($V_l = 0.3$ m/s; $V_g = 0.2$ m/s), stratified wave bubble flow ($V_l = 0.5$ m/s; $V_g = 0.4$ m/s) and annular wave bubble flow ($V_l = 1.0$ m/s; $V_g = 0.66$ m/s), along with the bubble flow ($V_l = 1.7$ m/s; $V_g = 0.53$ m/s) needed for subsequent investigations. The comparison results (Fig. 3) revealed consistent characteristics for each flow pattern in both the previous experiment and simulation. In the stratified bubble flow (Fig. 3(a)), the relatively low velocities of the gas and liquid resulted in substantial gas accumulation in the upper region between the inlet and element C1, forming a distinct gas-liquid interface. As the gas reached a critical accumulation level, large bubbles and a few small bubbles entered element C2 under the influence of varying velocities. Notably, bubble generation primarily occurred in elements C3 and C5 due to altered rotation direction, inducing increased fluid turbulence and vortex formation, with velocities reaching nearly 1.5 m/s, which facilitated bubble formation. Compared with the experiment, Zone D displayed a clear gas-liquid interface and a minimal amount of gas entering the liquid phase.

With further increases in the gas and liquid velocities, gas accumulation at the inlet diminished, and the phenomenon of gas upwelling and aggregation in C4 and C5 vanished (Fig. 3(b)). The stratified wavy bubble flow featured an apparent gas-liquid interface that emerged in Zone D, characterized by wavy profiles. The liquid phase contained a large number of bubbles, consistent with the experimental characteristics and distinguishing it from the stratified bubble flow. In the annular wave bubble flow (Fig. 3(c)), the elimination of gas upwelling at the inlet resulted from sufficiently high liquid velocities. However, the gas still exhibited an upward mobility tendency. Thorough mixing between the gas and liquid occurred in the static mixer, but owing to excessive gas volume, complete entrainment into the liquid phase was unachievable, leading to the formation of a large bubble in the central region of Zone D, surrounded by smaller bubbles progressing forward. This aligned with experimental observations.

Achieving a bubble flow was possible by controlling the suitable range of gas and liquid velocities ($V_l = 1.7$ m/s; $V_g = 0.53$ m/s), representing the ideal flow pattern required for equipment operation under real conditions (Fig. 3(d)). The gas flowed parallel to the pipe, and moved forward in the inlet section. In this case, the liquid velocity was adequate to rupture the continuous gas columns, generating fine and dense bubbles in Zone D, which is consistent with the experimental images.

Subsequent simulations were conducted under bubble flow conditions for further validation. The gas phase distribution and velocity profiles for various gas velocities (0.033–0.231 m/s) at a constant liquid velocity of 1 m/s are shown in Fig. 4(a). With increasing gas velocity, the length of the gas slug formed in Zone B increased, and more gas accumulated within the static mixer. At a gas velocity of 0.033 m/s, no noticeable accumulation was observed in the mixed gas. However, as the velocity increased to 0.099 m/s, gas accumulation occurred in the first part, and further increases resulted in greater gas accumulation at this location. When the velocity reached 0.165 m/s, accumulation occurred in the second part. These accumulation phenomena were primarily caused by changes in the flow streamlines due to the alteration of the

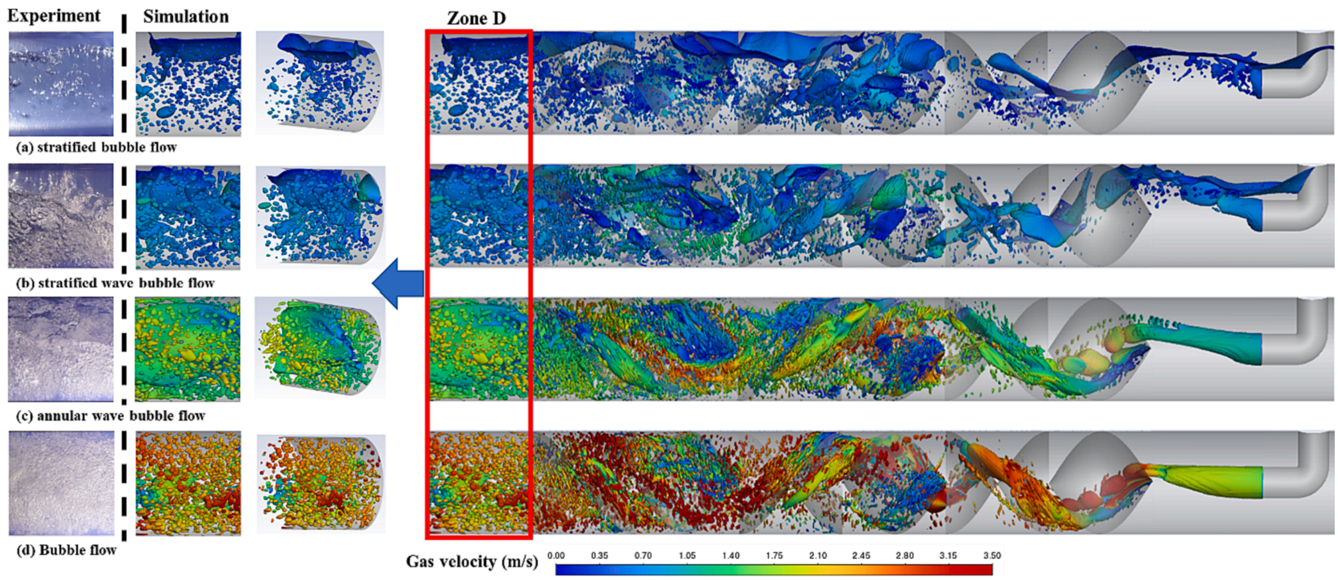


Fig.3. Identification of gas-liquid flow pattern from the simulation and experiment.

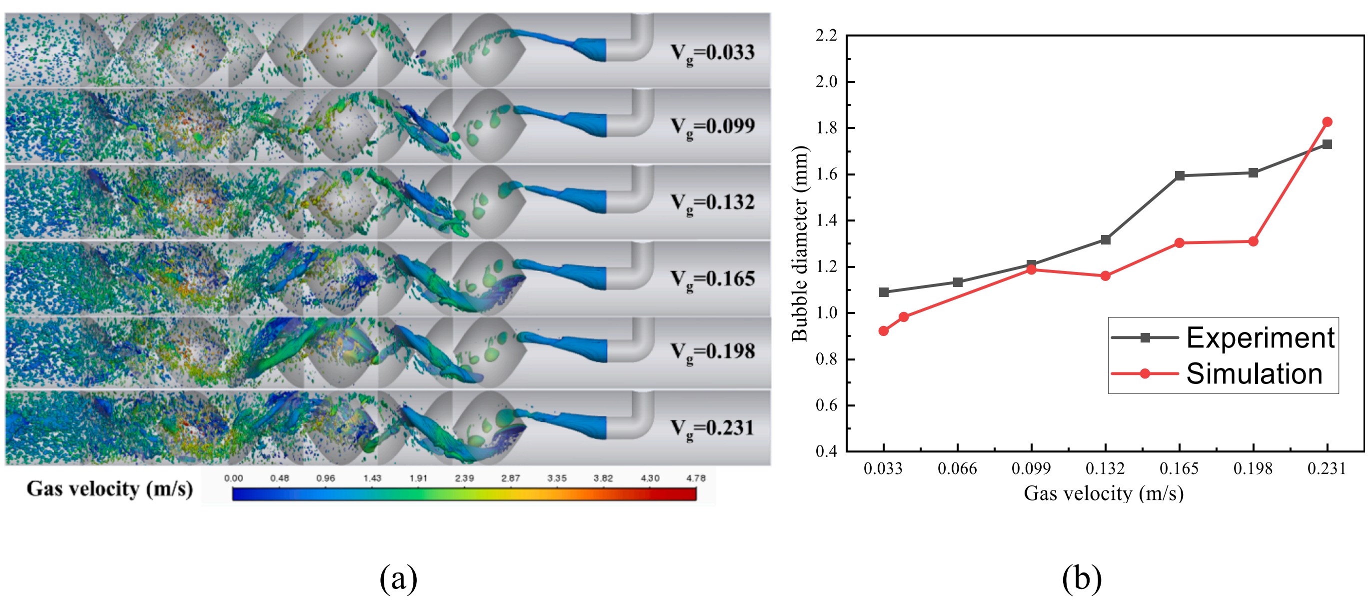


Fig.4. Gas distribution and velocity (a) and variation of mean bubble diameter in the experiment and simulation (b) with gas-phase flow rate.

mixing element. Additionally, it was evident that the bubble size in Zone D increased with increasing gas velocity. The impact of gas velocity on flow behavior within the static mixer was determined. Smaller bubble diameters can provide a larger specific surface area, thereby increasing the mass-transfer efficiency of the gas-liquid system [14,45]. As shown in Fig. 4(b), the mean diameter of the bubbles increased with the increase in the gas flow rate, and the experimental and simulated results exhibited a relatively small discrepancy.

3.2. Gas-liquid mixing mechanism analysis

The mixing elements of the static mixer are important components of the HiPOx^{TM} reactor and the core of gas-liquid mixing, which combines two or more different fluids through specific structural designs and fluid mechanics principles. A better understanding of the mixing mechanisms can help achieve improved gas-liquid mixing efficiency.

Numerical simulations were conducted for the first part of mixing elements with a gas inlet velocity of 0.27 m/s and liquid inlet velocity of 1.00 m/s; the temporal evolution of the gas distribution is shown in Fig. 5. When the gas-liquid flow in the circular tubes with and without the mixing elements were compared, it can be seen that the flow fields were consistent before the gas entered the mixing elements (0.1 s). Once the gas entered C1, the streamlines changed and stretched. The entire process within C1 involved the stretching and breaking of the gas slug into large bubbles. However, bubble aggregation and rotation occurred subsequently, leading to the formation of a gas slug. Between C1 and C2, further elongation of the gas slugs took place. Toward the front of C2, the gas slugs were predominantly divided into larger and smaller portions. Eventually, they decomposed into smaller bubbles and entered the next element. In the circular tube reactor without mixing elements, the gas slug floated upward and flowed forward to the next part. In general, the streamline was altered by the mixing elements in the first part, causing rotation and stretching of the gas, resulting in the formation of large bubbles.

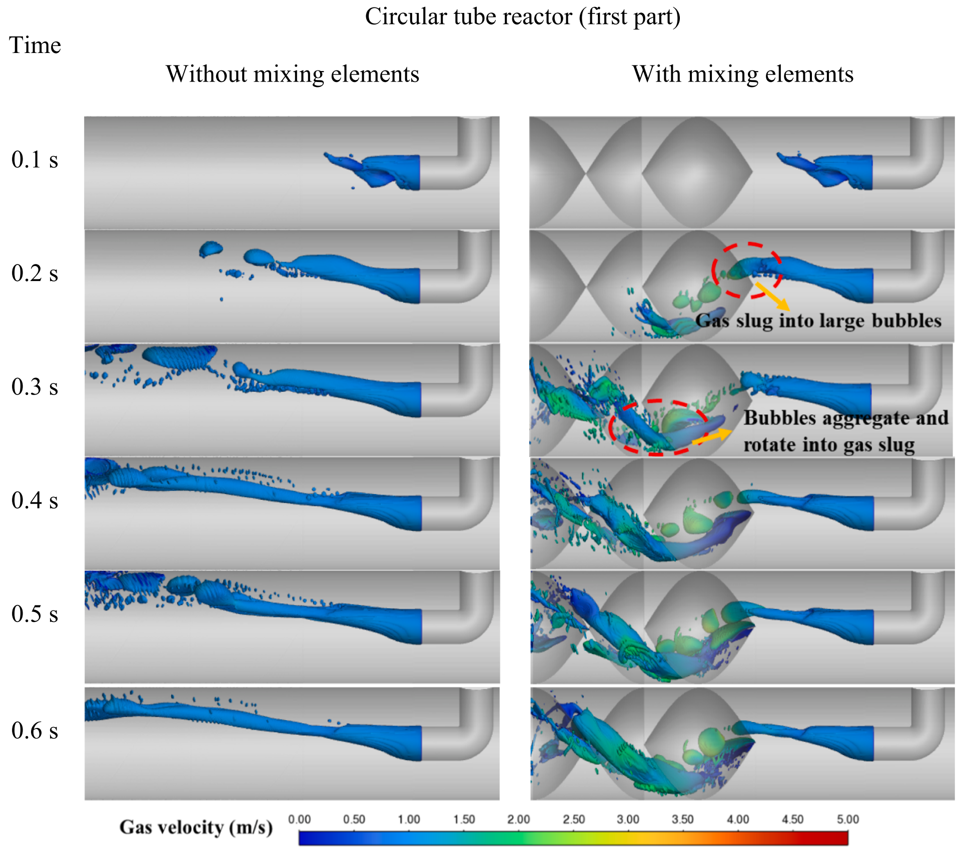


Fig.5. Temporal evolution of the gas in the first part.

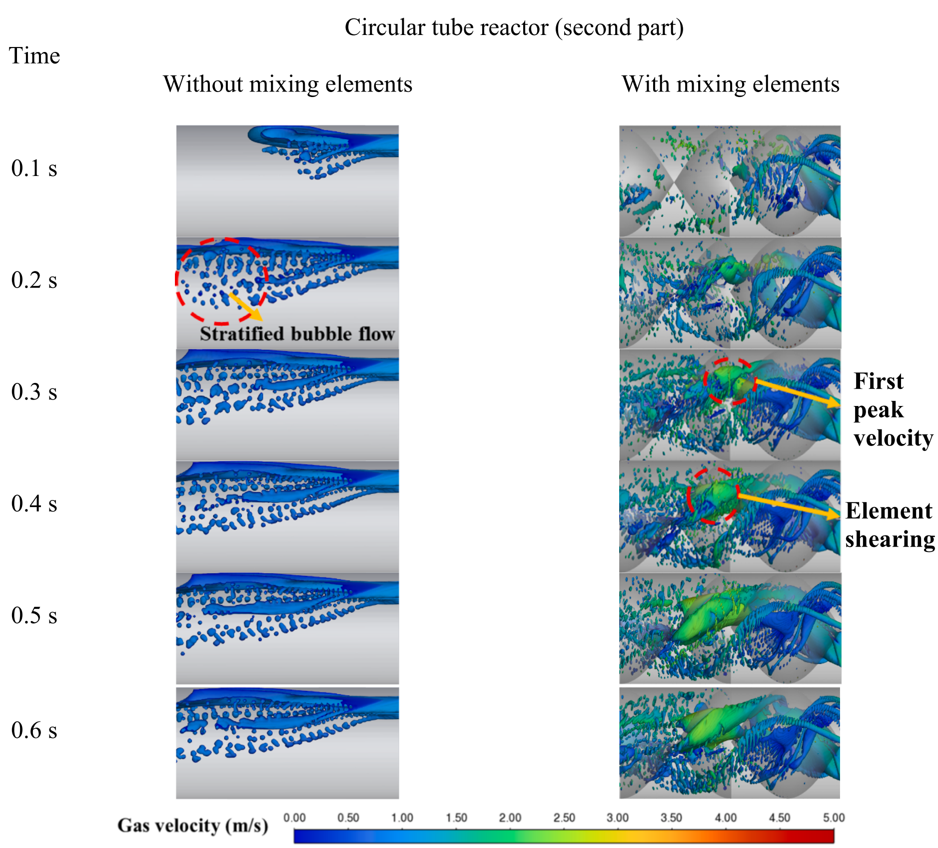


Fig.6. Temporal evolution of the gas at the second part.

As illustrated in Fig. 6, the gas slug that was originally supposed to enter from the upper section was transformed into two bubble groups at the inlet of the second part, dispersing small bubbles in the upper section and aggregating large bubbles in the lower section, as it entered C3. The upper portion underwent effective dispersion into small bubbles after passing through C3. Subsequently, the flow velocity increased because of the first direction change within C3.

In gas-liquid flows, the reduction in bubble size is primarily attributed to the increase in fluid velocity, which increases the inertial forces leading to shear bubbles. However, resistance to bubble deformation is governed by the surface tension of the gas-liquid interface [46]. To compare the influences of these factors, the Weber number was, defined as the ratio of inertial forces to surface tension, as follows:

$$We = \frac{\rho v^2 l}{\sigma} \tag{12}$$

where ρ is the density of the mixed phase, v denotes the fluid velocity, l represents the pipe diameter, and σ represents the surface tension. In general, an increase in We implies a larger impact of inertia forces relative to surface tension, resulting in smaller bubble diameters [47,48].

It was calculated that from elements C1 to C3, the We increased from 2201.43 to 2768.91. This indicated that the influence of the inertial forces relative to the surface tension increases as We moved from elements C1 to C3, resulting in a decrease in the bubble diameter. The substantial increase in We was primarily attributed to the increase in the fluid velocity. Upon entering C1, the fluid maintained a low-velocity state. However, as it progressed through the subsequent elements, the static mixer induced alterations in the fluid streamlines, enhancing the turbulence and resulting in an increase in the flow velocity [24]. The gas slug fractured and dispersed under the combined influence of the front part of element C4 and the increase in velocity. No significantly large bubbles were observed. In a circular tube reactor without mixing elements (left side of Fig. 6), the gas underwent a transition from a gas slug to a stratified bubble flow. Most of the gas tended to rise and form a distinct gas-liquid interface. However, a small fraction of the gas was sheared into small bubbles under the influence of the liquid phase and continued to flow forward. Therefore, the mixing elements could uniformly distribute the gas that originally formed stratified layers and induce fluid acceleration to increase the impact of the inertial forces, resulting in bubble dispersion.

In the circular tube reactor without the mixing elements, the stratified bubble flow pattern was sustained from the outlet of the second section. With the addition of the mixing element, significant accumulation of gas occurred, and the flow rate reached a maximum, forming a vortex (Fig. 7), owing to the effect of element C5. As the vortex grew and reached a critical point, it released some larger bubbles. Subsequently, these bubbles became smaller owing to fragmentation and stretching, until a stable flow of bubbles was formed in zone D. The gas velocity increased significantly at the positions where the flow direction changed within element C3 in the second part and C5 in the third part, and the We increased from 2768.91 to 3057.05, the re-enforcement of the impact of inertial forces leading to smaller bubbles. In the third part, under the influence of the second change in direction, the velocity increased and a vortex formed, leading to further dispersion and mixing of the gas phase.

In the first part, C1 and C2 induced a change in the fluid streamlines, leading to the rotation and stretching of the gas, resulting in the formation of larger bubbles. In the second part, elements C3 and C4 facilitated the uniform radial distribution of the gas and accelerated the fluid. This acceleration caused the impact of the inertial force on the bubbles to increase relative to the surface tension, leading to their dispersion. Finally, in the third part, elements C5 and C6 generated a vortex flow and secondary acceleration of the fluid, further promoting dispersion and mixing throughout the system. In summary, the mixing elements induced streamline variations between the two phases (Fig. 8), resulting in local acceleration and stretching of the fluid. After the fluid accelerated, the effect of the inertial force on the bubbles increased relative to the surface tension. This leds to fragmentation and dispersion of the bubbles. Therefore, the mixing elements were not the direct factors causing bubble dispersion. Their primary role was to facilitate distribution mixing.

3.3. Gas-liquid mixing evaluation

An assessment was carried out to evaluate the effect of the mixing element on the gas-liquid mixing efficiency and pressure drop as well as the influence of the number of elements (4, 6, 8, and 10). The gas-liquid mixing degree was evaluated on two scales, macro- and micro-mixing, using the relative standard deviation (RSD) and bubble diameter, respectively. The energy consumption in the mixer was evaluated using the pressure drop as a metric.

The RSD illustrates the alteration in the mixture compared with the initial unmixed state [49]. It provides a quantitative measure of the degree of mixing in the gas-liquid flow, calculated using Equation (11). This calculation captures the variation or dispersion of the concentration or composition of the mixture.

$$RSD = \frac{\sigma}{\sigma_0} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(C_i - \overline{C})^2}}{\sqrt{\overline{C}(1-\overline{C})}} \tag{13}$$

where σ and σ_0 are the standard deviations of the gas-phase volume fractions after mixing and in the initial unmixed state, respectively. $\overline{C}$ is the mean gas-phase volume fraction, N denotes the number of sampling points on the cross-sectional plane, and C_i is the gas-phase volume fraction at point i.

RSD serves as an indicator for evaluating the effectiveness of the static mixer in achieving the desired level of mixing [25]. It assesses the overall mixing performance and uniformity of the mixture, and is sensitive to changes in the mixing efficiency of the static mixer. A low RSD indicates a high level of mixing uniformity and efficiency. In addition, by utilizing UDF to extract the bubble diameter distribution, the arithmetic mean diameter was calculated from the obtained data.

The variations in the RSD and bubble diameter with axial distance for the four elements are illustrated in Fig. 9(a). The RSD continuously increased within the elements until Zone D, where it started to decrease, and the continual reduction in bubble diameter could be because the gas entered element C1 in the form of larger gas bubbles, which then coalesced and stretched in element C2 to form slightly smaller gas bubbles before entering element C3. With a change in the flow direction within the unit, the gas bubbles further aggregated until they reached a certain extent of coalescence. Subsequently, because of the increased velocity, they stretched to the midsection of element C4, where they dispersed and transformed into bubbles entering Zone D.

The RSD and bubble diameter variations in the first four elements under elements 6, 8 and 10 follow a similar pattern (Fig. 9(b – d)). Furthermore, the RSD decreased significantly at element C5, which could be attributed to the generation of eddies at this location, leading to a more uniform radial distribution of the gas phase. The continuous decrease in the RSD of the six elements beyond C5 indicated that the mixing effect of this element persisted for a certain distance after leaving the static mixer. However, the slight increase in the bubble diameter after reaching Zone D was due to the coalescence of a small number of bubbles in the absence of subsequent elements. For eight elements, there was a minor decrease in bubble diameter after six elements. The RSD of 8 and 10 elements showed a slight increase at element C8, which was also caused by the clustering of gas owing to the change in flow direction. However, the increase in bubble diameter after C9 in 10 elements was likely due to insufficient pressure at this location to continue the breakup and dispersion of the bubbles.

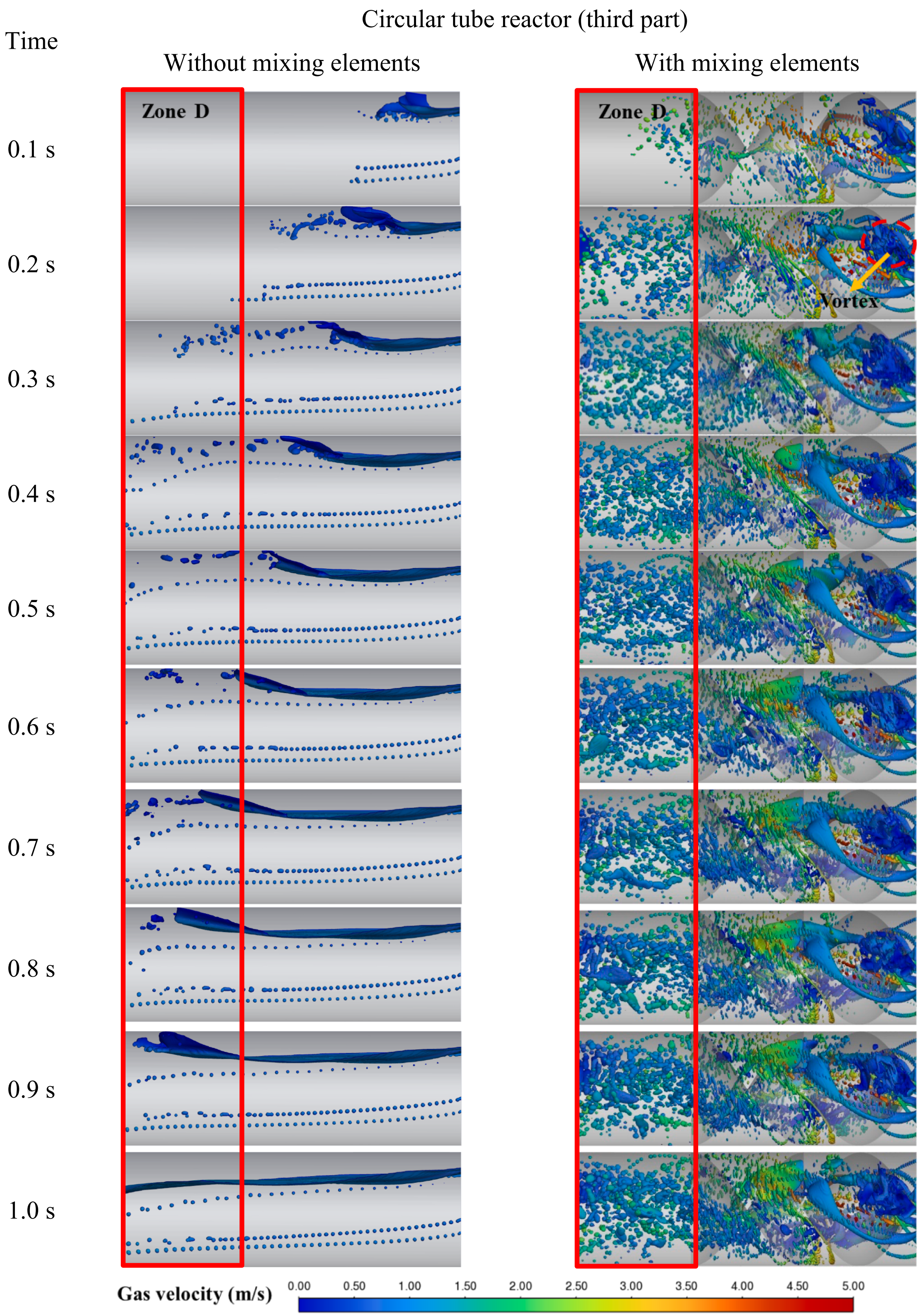


Fig.7. Temporal evolution of the gas at the third part.

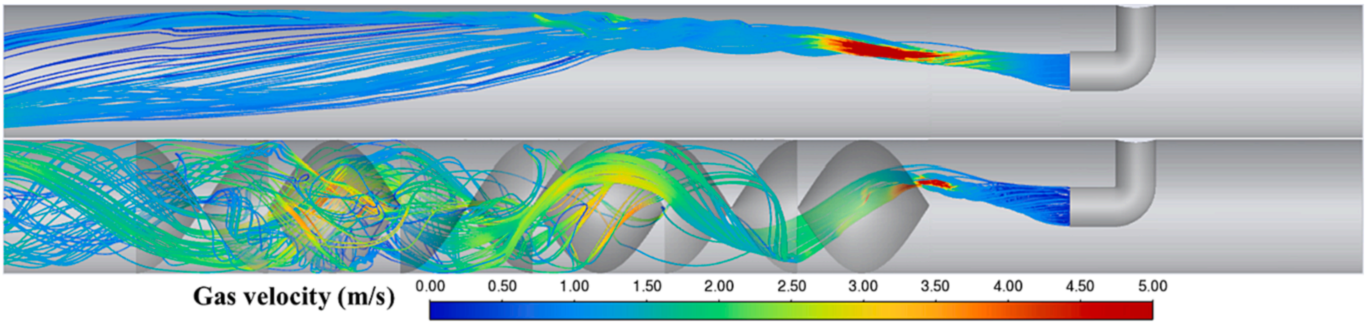


Fig.8. Overall streamline with and without mixing elements.

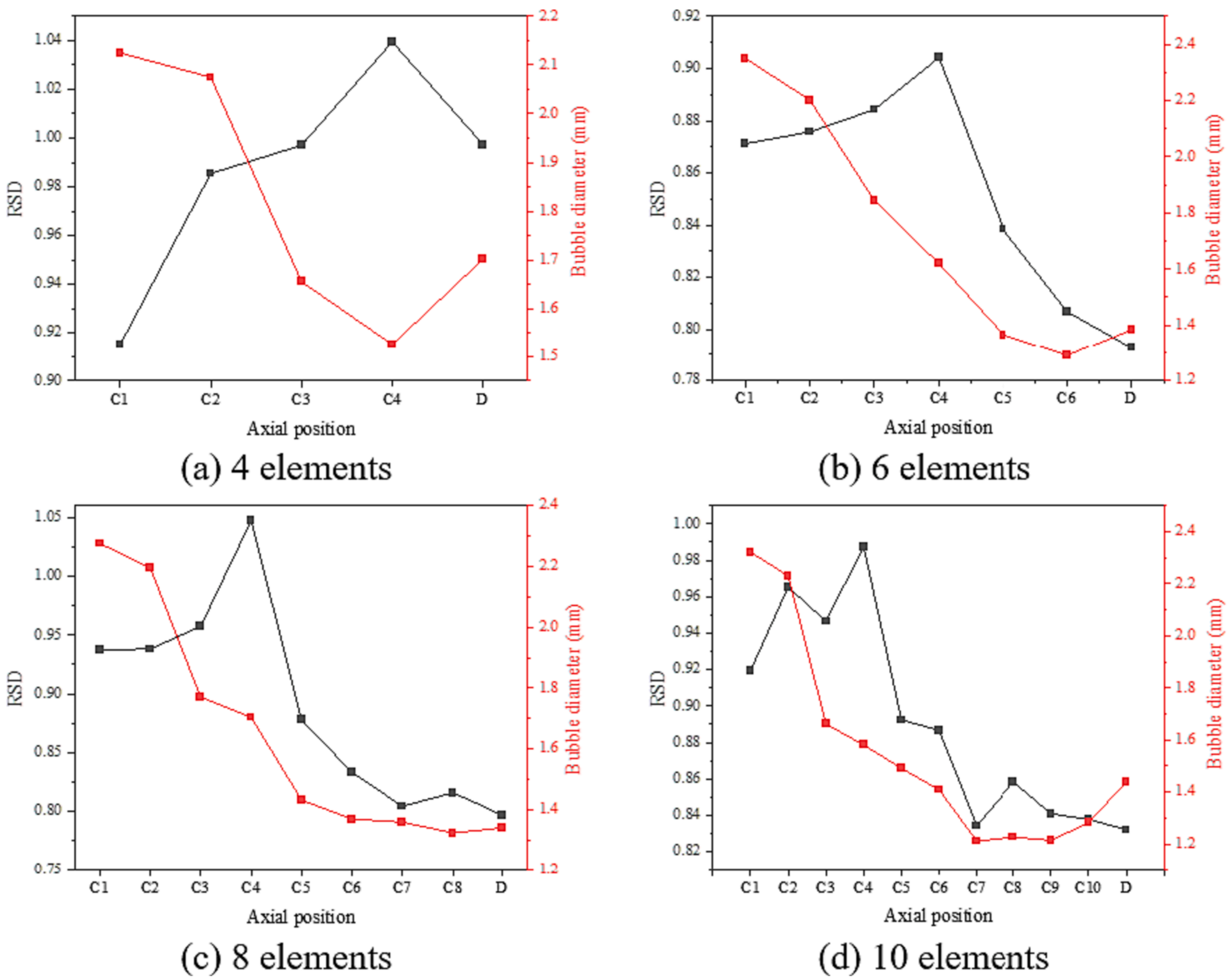


Fig.9. The RSD and bubble diameter versus the number of elements.

As shown in Fig. 10, the bubble diameter and RSD in Zone D vary with the number of elements and pressure drop. When the number of elements decreased to four, the RSD increased from 0.793 to 0.997 and bubble diameter increased from 1.384 to 1.702 mm, representing increases of 25.7 % and 23.0 %, respectively. The degree of mixing in Zone D significantly decreased. Even if the pressure drop decreased from 6918.45 to 3980.32, a decrease of 42.5 % was unreasonable. When the number of elements was increased to eight, the RSD increased from 0.793 to 0.797 and bubble diameter decreased from 1.384 to 1.340 mm, representing an increase of 0.5 % and decrease of 3.2 %, respectively. The degree of mixing in Zone D did not show any significant changes. However, the pressure drop increased from 6918.45 Pa to 9520.71 Pa, indicating a significant increase of 37.6 %. The RSD, bubble diameter, and pressure drop with 10 elements were higher than those with six elements.

4. Conclusions

To analyze the gas-liquid flow field and evaluate the mixing mechanism of the static mixer, a static-mixer-based plug-flow reactor derived

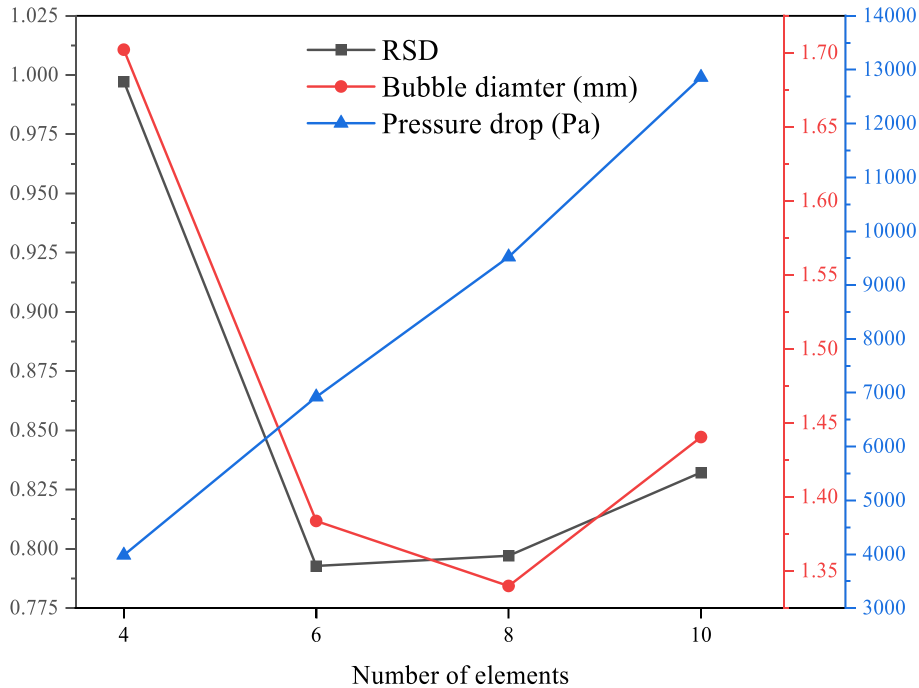


Fig.10. The RSD, bubble diameter and pressure drop variations in Zone D.

from the HiPOx^{TM} reactor system was investigated by CFD methods. The plug-flow reactor was divided into three parts, each containing two mixing elements. The bubble evolution inside the reactor and the gas dispersion and gas-liquid mixing processes were analyzed and identified. The conclusions were as follows.

Three composite flow patterns—stratified bubble flow, stratified wave bubble flow, and annular wave bubble flow—were observed in both simulation and experimental results. The influence of gas velocity on bubble size was analyzed. And it found that as the gas velocity increased, the size of the accumulated bubbles also increased. Based on the verification results, it was determined that the mixing element primarily promoted mixing, rather than gas dispersion by comparing the gas-liquid flow fields with and without the static mixer. The mixing elements induced streamlined variations between the two phases, resulting in the local acceleration and stretching of the fluid. This acceleration of the fluid increased the impact the inertial force on the bubble relative to the surface tension, leading to bubble fragmentation and dispersion.

The RSD and bubble diameter were used to evaluate the gas-liquid mixing degree at two scales: macro- and micro-mixing. The plug-flow reactor equipped with six mixing elements achieved good mixing performance (RSD of 0.793 and bubble diameter of 1.384 mm) with the lowest energy consumption. Its RSD decreased by 25.7 % and bubble diameter decreased by 23.0 % compared with the reactor with four elements, whereas the energy consumption was 37.6 % lower than that of the reactor with eight elements.

CRediT authorship contribution statement

Lvliang Wang: Writing – original draft, Data curation, Formal analysis, Writing – review & editing. **Yihan Peng:** Data curation. **Xuejing Yang:** Conceptualization, Funding acquisition, Project administration. **Yuanyuan Qian:** Funding acquisition, Project administration. **Hualin Wang:** Funding acquisition, Project administration. **Yanjing Xu:** Funding acquisition. **Yanxia Xu:** Funding acquisition, Methodology, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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